66. In which of the following are hydrogels used?
1. Controlled drug delivery in patients
2. Mobile air-conditioning systems
3. Preparation of industrial lubricants
Select the answer using the code given below:
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans.(d):
- A hydrogel is a water-based material with a soft consistency and moisturizing properties.
- Hydrogels are used in the biomedical industry for therapeutic delivery. It is also used in tissue engineering. In ophthalmology, intraocular lenses, Contact lenses and corneal prostheses, bone cement for orthopedics, Hydrogels are also used in wound dressings.
- **Hygiene products**–Hydrogels are used in diapers and sanitary pads as they can absorb high amounts of liquid.
- **Construction-**Hydrogels are used to retain moisture in concrete, improving its setting process.

– These properties and uses of hydrogels indicate its important role in various fields.

67. Which one of the following is the exhaust pipe emission from Fuel Cell Electric Vehicles, powered by hydrogen?
(a) Hydrogen peroxide (b) Hydronium
(c) Oxygen (d) Water vapour

Ans.(d):
- The emissions from hydrogen-powered fuel cell electric vehicles (FCEVS) are mostly water (H_2O). The chemical reaction between hydrogen and oxygen in a fuel cell produces electricity, and the only by products of this process are steam and hot air. Therefore, these vehicles do not emit harmful gases, making them much more environmentally friendly than conventional petrol or diesel powered vehicles.

68. Recently, the term ''pumped-storage hydropower'' is actually and appropriately discussed in the context of which one of the following?
(a) Irrigation of terraced crop fields
(b) Lift irrigation of cereal crops
(c) Long duration energy storage
(d) Rainwater harvesting system

Ans.(c):
- Pumped storage hydropower (PSH) is a type of hydroelectric energy storage.
- PSH acts like a giant battery, because it can store electricity and then release it when it is needed.
- PSH is a configuration of two water reservoirs at different elevations that can generate electricity as water moves down (discharge) from one to the other, passing through a turbine.

69. ''Membrane Bioreactors'' are often discussed in the context of:
(a) Assisted reproductive technologies
(b) Drug delivery nanotechnologies
(c) Vaccine production technologies
(d) Waste water treatment technologies

Ans.(d):
- Membrane bioreactors are discussed as waste water purification technologies.
- Membrane Bioreactors (MBR) are a modern water and waste water purification technology that integrates biological treatment and membrane filtration.
- It consists of the following main elements–
Biological reactor–Where micro organisms decompose waste.
Membrane unit–Which performs solid-liquid separation, producing high-quality clean water.
- It can be used for domestic waste water treatment, industrial waste water treatment and production of reusable water.
- **Working Mechanism**–MBR systems first undergo conventional biological purification, in which micro organisms decompose organic material, next, a membrane filtration unit separates clean water by filtering out solid particles. The membrane is usually micro filtration or ultra filtration.

70. With reference to the Indian economy, ''Collateral Borrowing and Lending Obligations'' are the instruments of:
(a) Bond market (b) Forex market
(c) Money market (d) Stock market

Ans.(c):
- Collateralized borrowing and lending obligation (CBLO) is a money market instrument that represents an obligation between the borrower and the lender regarding the terms and conditions of the loan.
- It is mainly used for short-term money management.
- CBLO is operated by clearing corporation of India limited (CCIL) and Reserve Bank of India (RBI).
- This instrument is available to banks, primary dealers, mutual funds, insurance companies and other financial institutions who use it for liquidity management and meeting cash requirement.
- The purpose of CBLO is to facilitate short-term borrowing and lending facility to banks and other financial institutions without any risk. Under this, both the borrowing and lending parties use eligible securities as collateral through The Clearing Corporation of India Limited (CCIL).

71. The organisms ''Cicada, Froghopper and Pond skater'' are:
(a) Birds (b) Fish
(c) Insects (d) Reptiles

Ans.(c):
- **Cicada**–A cicada is a **type of insect** that is part of the cicadoidea family. These insects are known for their distinctive sound, which they produce mainly during mating time.
- Cicadas are important to the ecosystem because they improve soil quality and help cycle nutrients among vegetation.
- **Froghopper**–is a **small insect** also known as 'spittlebug'. It is known for its jumping ability, it is able to jump many times the distance of its body length. Froghopper belongs to cercopoidea family
- This insect depends on the juice of plants and sometimes it is also harmful for the farmers.
- **Pond skaters**–are commonly found in ponds, rivers and quiet reservoirs. They are also known as water striders. They are a **type of insect** belonging to the Gerridae family. They are known for their ability to walk on the surface of water. This feat is achieved due to their long, hydrophobic legs that distribute their weight and allow them to take advantage of the surface tension of the water.

72. Consider the following statements:
Statement-I: Many chewing gums found in the market are considered a source of environmental pollution.
Statement-II: Many chewing gums contain plastic as gum base.
Which one of the following is correct in respect of the above statements?
(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
(c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

Ans.(a):
- Chewing gum is a soft, sticky substance that can be chewed but not swallowed. It is primarily composed of gum base, sweeteners, flavorings and softeners.
- Fillers like talc and calcium carbonate help increase the size and bulk of the gum, polymers such as polyvinyl acetate (PVA) give the gum its elasticity and emulsifiers are used to add flavors and colors to the gum and reduce stickiness, softeners such as vegetable oils are added to the gum base to keep it chewable and soft.
- Regular chewing gum contains a substance called polyvinyl acetate (PVA) which is used to make tires and glue. The gum we eat is plastic based. Research Gate reported that chewing gum produces more than 105 tons of plastic waste every year. Thus, the discarded non-biodegradable remnants of gum create plastic pollution.

73. Consider the following pairs:

	Country	Animal found in its natural habitat
1.	Brazil	Indri
2.	Indonesia	Elk
3.	Madagascar	Bonobo

How many of the pairs given above are correctly matched?
(a) Only one (b) Only two
(c) All three (d) None

Ans.(d):
- **Indri**–Indri, also known as babakoto. Indri is a type of lemur found in the eastern rainforests of Madagascar. Indri is famous for its loud, terrifying call which can be heard from for away. Indri is considered an endangered species, its numbers are decreasing due to habitat destruction, hunting and continuous reduction in forest area.
- **Elk** –The Elk (Cervus Canadensis) is one of the largest deer family (cervidae) in the world, and one of the largest land mammals in North America and East Asia.
- **Bonobo**–The Bonobo is an endangered ape found in the forests of Central Africa. The Bonobo has also been historically known as the 'pygmy chimpanzee'.
- The Bonobo is found in an area of 500000 square km within the Cango Basin of the Democratic Republic of the Congo (DRC) in Central Africa.

74. Consider the following statements regarding World Toilet Organization:
1. It is one of the agencies of the United Nations.
2. World Toilet Summit, World Toilet Day and World Toilet College are the initiatives of this organization, to inspire action to tackle the global sanitation crisis.
3. The main focus of its function is to grant funds to the least developed countries and developing countries to achieve the end of open defecation.
Which of the statements given above is/are correct?
(a) 2 only (b) 3 only
(c) 1 and 2 (d) 2 and 3

Ans.(a):
- The World Toilet Organization (WTO) was founded by Jack Sim on 19 November, 2001 in Singapore with the Goal of improving toilet and sanitation conditions around the world. It is a non-profit organization (NGO). Hence, the WTO (World Toilet Organization) is not a United Nations Agency.
- With the establishment of the WTO on 19 November, 2001, it was announced to celebrate this day as 'World Toilet Day', which received official recognition of 'United Nations Day' in the year 2013.
- The WTO works to improve sanitation facilities globally through various initiatives, campaigns and partnerships, especially in areas where lack of sanitation causes serious health problems.
- Programs like 'World Toilet Summit' and 'Urgent Run' are organized by the WTO. Through the Urgent Run Program, immediate action is called for to deal with the sanitation crisis.
- The World Toilet College was launched in 2005 by the WTO as a social enterprise aimed at promoting innovative research in toilet design, sanitation and hygiene technologies, and ensuring hygiene standards.
- Other initiatives of the organization are the Rainbow School Toilet initiative in 2015 and the 'floating community Toilet' Project in 'Tonle Sap Lake', Cambodia in 2018.
- Its main function is not to provide funds to LDCs and developing countries.
- The WTO was granted consultative status with the United Nations Economic and Social Council in 2013.

75. Consider the following statements:
1. Lions do not have a particular breeding season.
2. Unlike most other big cats, cheetahs do not roar.
3. Unlike male lions, male leopards do not proclaim their territory by scent marking.
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans.(a):
- Lions do not have a specific breeding season. Usually, it can occur throughout the year.
- Cheetah do not roar, cheetahs are not capable of roaring like other big cats. Cheetahs belong to the "Purring Cats" subfamily and as such do not roar. On the other hand, animals like lions and tigers are capable of roaring.
- Male leopards actually declare their territory by scent marking. They mark their territory in various ways, among which scent marks are prominent. They spray urine, scratch trees with their claws and sometimes even use feces to scent mark. These scent marks inform other leopards about the boundaries of their territory and discourage any encroachment.
- In contrast, lions also mark their presence and territory by scent marking but lions usually live in herds and defend their territory. Collectively, while leopards live alone and defend their territory individually.
- The International Big Cat Alliance (IBCA) is a mega global alliance launched by India during the 50th anniversary of Project Tiger in April, 2023. The alliance aims to conserve the world's seven major big cats, which include the Tiger, Lion, Snow Leopard, Leopard, Jaguar, Puma and Cheetah.

76. Which one of the following is the correct description of ''100 Million Farmers''?